costing_CONFIDENCE

Confidence ratings (ordering and works-as-intended)

Two columns for every project. They answer different questions:

- **Order confidence:** how sure we are that you can buy exactly these parts and files and get what the BOM says. **High**: every LCSC part live-checked (identity, package, stock above 500, not end-of-life) and the Gerber/BOM/CPL files pass all automatic gates. **Medium-High**: same, but one or more parts have limited stock (still enough for a prototype run) or a substituted part whose pad geometry is unproven. **Medium**: parts depend on a vendor module, a hand-sourced item or a manual cross-check. **Low**: no board or no verified parts yet.
- **Works-as-intended confidence:** how sure we are that the design does its job when built. An engineering judgement, **not a measurement**: nothing has been built. It weighs (a) how standard the circuit is (datasheet reference design versus novel), (b) the evidence we have (host-run firmware tests, simulation, pin-map checks, datasheet equations), (c) risks that only hardware can reveal (analog front ends, RF, power switching, mains, calibration, uncompiled firmware). **High** would need a built and tested unit, so no project is rated High. **Medium-High**: standard circuit, strong test evidence, low-risk failure modes. **Medium**: expect to find one or two issues in bring-up. **Low-Medium**: significant parts of the behaviour are assumed. **Low**: concept or software only.

Plan for one respin. Even a clean design review usually leaves one or two bring-up findings on a first board; order the small quantity first (5 boards) and keep the files in this repository as the single source of truth.

Project	Order confidence	Works-as-intended	Why (works)	First thing to do
Seeti Counter	**Medium-High** (limited stock on: BZ1 Buzzer 3V (stock 140))	**Medium**	Mic front end and clap/whistle detector are analog and tuned on synthetic audio; the firmware itself passes 115 host checks; the documented limit is that quiet whistles are missed.	Bench-test the microphone gain with real claps before ordering more than 5 boards.
Gas Taraju (LPG cylinder scale)	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium**	HX711 load-cell chain is a standard circuit; accuracy depends on the cell, mounting and calibration, which only a build can show.	Calibrate with a known weight; check drift over a day.
Fridge Door Guard	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140); D4 TVS 5V bidir (stock 175))	**Medium-High**	Reed switch, timer and buzzer: simple circuit, simple firmware; TVS and input conditioning follow standard practice.	Test the reed gap and the 2-minute alarm on the bench.
Bijli Meter Reader	**High** (all parts live-checked, none flagged)	**Medium**	Optical pulse reading of a meter LED depends on the sensor alignment and ambient light; circuit is standard.	Try on a real meter with a shroud before enclosure work.

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Bijli Log	**High** (all parts live-checked, none flagged)	**Medium**	Current/voltage logging front end is analog and uncalibrated; mains-adjacent measurement needs isolation care.	Do not connect to mains until a qualified electrician reviews the isolation; start with a bench supply.
Chulha Timers	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium-High**	Multi-timer with buttons, LEDs and buzzer; simple circuit; firmware tested on the host.	Check button debounce and the buzzer volume on the first board.
Rain Alert Bell + plate	**Medium-High** (limited stock on: BZ1 Buzzer 85dB (stock 140); D3 TVS 5V bidir (stock 175); D4 TVS 5V bidir (stock 175))	**Medium**	Conductive rain plate corrodes and drifts; electronics are simple and protected with TVS.	Use the plate for a week outdoors before deciding on a coating.
Tap Litre Meter	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140); D4 TVS 5V bidir (stock 175))	**Medium**	Hall/flow-pulse counting is standard; the litres-per-pulse constant depends on the flow sensor.	Calibrate against a measured bucket.
Masala Rack (laser-cut)	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium-High**	Laser-cut rack: fit depends on kerf; stability limit stated.	Cut a kerf coupon first.
Laptop Stand (laser-cut)	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium-High**	Laser-cut stand: fit depends on kerf; stability limit stated.	Cut a kerf coupon and load-test.
1 Desi Games Studio (print)	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Printed game; needs language and fact review before print.	Proofread with native speakers.
2 Kalakriti Automata (laser)	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Laser cut files regenerated with correct kerf compensation (outlines outward, holes inward, gear backlash only once); cam and gear checks PASS; frame layouts and figure artwork are still first-pass.	Cut one kit in scrap MDF and run the mechanism.
3 Prayog STEM lab kit	**High** (all parts live-checked, none flagged)	**Medium-High**	Educational kit with a straightforward MCU board; 40 host checks on the statistics and command code.	Run the kit's own experiments on the first unit.
4 Jal-Rakshak tank controller	**Medium-High** (limited stock on: J1 12 V DC in (stock 321))	**Medium**	Low-voltage board reuses the Kheti-Mitra 12 V front end and pump driver; the firmware rules pass 20 simulated checks and an 8 MHz PCB build exists (3.2 kB); real probes, water, pump currents untested; the mains version is still out of scope.	Bench test with a 12 V pump and probes in a bucket; never connect mains.
5 Kheti-Mitra irrigation	**Medium-High** (limited stock on: J1 12 V DC in (stock 321))	**Medium**	Irrigation controller with 12 V switching and soil sensing; 22 host checks pass; field conditions (water, sun, surge) are the risk.	Run with a lamp as the load for 48 hours before any valve or pump.

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6 Gas-Suraksha leak alarm	**High** (all parts live-checked, none flagged)	**Low-Medium**	Prototype/logging board carries the tested alarm engine (34 checks) with an active buzzer and a UART log header; MQ-6 thresholds are placeholders and it is not a certified detector.	Log real sensor data for a week; never use or sell as a safety device.
01 Reflex Raja	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium-High**	Shares the Desi Arcade board; game logic tested with about 80,000 host checks.	Play-test the buttons and the buzzer.
02 Taal Memory	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium-High**	Same board and test base as Reflex Raja.	Play-test the buttons and the buzzer.
03 Aloo Bomb	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium-High**	Same board; game logic host-tested; the potato-clock mechanics are the unknown.	Test timing with a real potato.
04 Beat Runner	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Medium-High**	Rhythm game on its own board; logic host-tested; LEDs and buzzer circuits are standard.	Check the LED brightness and button spacing.
05 Rangoli Spiro Gears	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium-High**	Laser-cut gears; kerf test first.	Cut a kerf coupon.
06 Paper Rocket Launcher	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	v1 changed to a stomp launcher (no pump, valve or stored pressure); laser-cut stand and parts list exist; range and energy are model estimates until measured.	Build one, fire 20 rockets in an open area and measure the range; keep foam noses mandatory.
07 Tilt Maze	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Gimbal now drawn (two sheets, kerf-compensated, tilt and clearance checks PASS) but never cut; dowel fits and knob feel need a test.	Cut one set in scrap MDF and test the tilt range and axle fits.
08 Rubber-band Racer	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium-High**	Cut file now kerf-compensated (wheels, axle holes, band slots at nominal size); simple mechanism.	Cut one in cardboard or MDF and test the band drive.
09 Marble Cricket	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	Cut file now kerf-compensated (peg holes at 6.2 mm press fit); pocket scoring needs a calibration drop.	Cut one, drop 200 marbles and check the pocket distribution.
10 Katha Cubes	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium-High**	Cube net and the 36 words ready; Hindi and Telugu checked for consistency (grandmother term aligned, gender-neutral problem phrases) but a native-speaker review is still required before print; icons are optional for the prototype.	Print text-only cubes for the prototype; get the words reviewed by native speakers before the production print.
KINETIC motion controller + vibration puck	**High** (all parts live-checked, none flagged)	**Medium**	ESP32-S3 controller with three StepStick sockets and a 24 V to 5 V buck. Motion library (370 checks) and protocol are host-tested; the ESP32 firmware now compiles for the ESP32-S3 with 0 warnings but has not run on hardware; drivers and motors are bought in. The vibration model is trained on synthetic data only.	Power with a current-limited supply and no drivers; then one driver and one motor at low current.

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MITTI-NETRA soil + structure node	**High** (all parts live-checked, none flagged)	**Medium**	Charger thermal pad now grounded with thermal vias; NTC window stops Li-ion charging outside about 6 to 44 C; power diodes corrected; IP65 box with a printed mounting plate; firmware core 23 checks and the pin map match; the cellular modem is a plug-in breakout and the ESP32 firmware compiles for the ESP32-S3 with 0 warnings (not yet run on hardware).	Bring up power and charging first (check TEMP voltage at room temperature), then the modem last.
BioTray vertical-farm controller	**Medium-High** (limited stock on: J2 12 V DC in (stock 321))	**Medium**	12 V low-side drivers, ESP32 and sensors; 22 host checks; calibration constants are placeholders; wet environment.	Dummy loads on all outputs first; seal the enclosure before any water.
BioWatt MFC harvester	**Medium-High** (all parts verified; the BQ25570 stock is limited (about 400) and the hold capacitor is a substitute grade)	**Low-Medium**	BQ25570 with megohm setting dividers computed from datasheet equations; cold start needs at least 600 mV (marginal for one microbial fuel cell); the hold capacitor is X7R where the datasheet asks for low leakage; leakage on a dirty board shifts the thresholds.	Test with a bench supply at 0.8 V first, clean flux thoroughly, then try the fuel cell.
GEONODE foundation monitoring	**Medium** (uses other boards plus bought-in modules; each module's pinout must be checked against its datasheet)	**Low-Medium**	Detector logic and feature code are host-tested (54 checks) but on synthetic vibration; the crack signature is an assumption; node hardware is the MITTI-NETRA board with a LoRa module.	Record real vibration for a week and re-derive the thresholds before trusting any alarm.
AQUAPONICS controller	**Medium** (uses other boards plus bought-in modules; each module's pinout must be checked against its datasheet)	**Low-Medium**	Control and forecasting core passes 13 checks; biological limits are placeholders; no ammonia or nitrite sensing; live animals depend on it.	Run a week with water only; tune limits with an aquaculture expert before any fish.
SORTER bio-robotic platform	**Medium** (uses other boards plus bought-in modules; each module's pinout must be checked against its datasheet)	**Low**	Tracking core passes 28 checks; the vision is a colour-rule baseline tested on synthetic discs; the gripper geometry is untested; machine guarding is your responsibility.	Image real produce under fixed light and replace the classifier before building the conveyor.
AGROVER solar rover	**Medium** (uses other boards plus bought-in modules; each module's pinout must be checked against its datasheet)	**Low-Medium**	Navigation and power core pass 19 simulated-field checks; dead-reckoning only; chassis plates untested with real motors; needs a safety review and an emergency stop.	Drive on blocks first, then slowly in a clear area with the stop in hand.
Jal-Netra (engineering marvel)	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Low-Medium**	Clamp-on ultrasonic transit-time measurement has a hard analog front end; the manual itself says Gate 1 acoustic test comes first and it is not sellable.	Run the Gate 1 acoustic test with the evaluation setup before any more boards.
ASTRADAND gesture wand (sci-fi project)	**Medium** (substitute addressable LED whose pad geometry is unproven; other parts verified)	**Medium**	IMU gesture recognition validated against a Python reference (46/46) and on synthetic gestures; hardware (haptics, six addressable LEDs) is new; the LED part was replaced with a pin-compatible one whose pads are unproven.	Order 2 boards, check the LED footprint fit, then run the gesture self-test.

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Gravity toys (5)	**Medium-High** (no electronic parts to source: local laser-cutting or printing)	**Medium**	All five toys now have prototype drawings (Santulan lathe drawing, marble-tower posts and groove spec, ramp-racer car and ladder stand, Galton assembly guide, Drop Quest v1 rules with 60 solver-verified levels and laser parts); physics checks PASS; none built or play-tested.	Cut or turn one of each and play-test; measure ramp times and marble behaviour.
TRISHNA urinal hydration monitor	**Medium-High** (limited stock on: J1 12 V DC in (2 A) (stock 321); U2 AP63203WU-7 (stock 195))	**Low-Medium**	Controller is a standard ESP32 carrier (buck, H-bridge, MOSFET + flyback, TIAs); firmware compiles for the ESP32 and 19 host tests cover the frame, filters, EC maths and every state-machine path; CAD solids give a 2.02 mL chamber. Unproven until built: the printed flow cell must be leak-free, the bipolar GPIO EC drive and the photodiode gains need tuning on the bench, and the SG estimate from conductivity + colour is only as good as the per-user refractometer fit.	Print and pressure-test the sensor block, then run validation V1-V4 with the synthetic-urine fluids before any real use.
Wave 1 electronics toys	**Medium-High** (limited stock on: BZ1 Buzzer (stock 140))	**Low-Medium**	AP1 board added (ATmega328P, MP3 module socket, bought-in 4-digit 74HC595 display module on a 5-way lead, 7-LED ball lane on a 74HC595, 7-button resistor ladder, lamp and motor drivers); all three apps now build for the board (about 6 kB each) and pass 76 host checks. The MP3 command codes are from memory and must be checked on the real module; the ladder thresholds are calculated, not measured.	Bench-test one board per product: ladder readings, MP3 commands, display module polarity (one compile flag each).
Prototype packs (10)	**Medium** (breadboard parts and modules from local vendors (prototype budget lists them))	**Medium**	Bench prototype packs on breadboards with 26 host checks; no PCB.	Use as learning kits; do not sell.
DJ / party system	**Medium** (breadboard parts and modules from local vendors (prototype budget lists them))	**Low-Medium**	The full product is a 12-18 month programme; its phase-0 prototypes exist in `prototypes/` (D1 beat-synced light controller, D2 DJ MIDI controller on Arduino Nano: firmware compiles, filter/encoder logic host-tested; Mixxx mapping well-formed with 17 valid controls and no MIDI clashes, not yet tried in Mixxx).	Build D1 and D2 from the prototype guide and test with Mixxx and a speaker you already own.